Sonification of Turing Patterns Modeling Limb Development

Garrett Compston, Soha Ben Tahar, Victor Zappi, Sandra Shefelbine

Northeastern University,
Department of Music,
Boston, MA, USA
compston.g@northeastern.edu

ABSTRACT

Alan Turing's reaction diffusion equations are used in the field of mechanobiology to model the formation of patterns in the natural world [1]. It is theorized that these equations might also be used to model the development of limbs, observing the patterns present in bone structure, and applying these to growing domains. This work is somewhat inaccessible to the layperson, being rooted deeply in pure math, and with visual simulations often not being fully explanatory in what exactly is happening mathematically. We have attempted to bridge this gap in understanding and access through the process of sonification. The main connection point we have selected between the two fields is in the similarity of the mechanical string equation to the reaction diffusion equations. In brief, we are performing additive synthesis and selecting overtones in accordance with the so-called "modes" found in Turing's equations. We then take this concept and expand it to growing domain systems, finding mappings with the performance of linear stability analysis on these systems. What we have created are timbral mappings of several simulations of Turing's reaction diffusion equations with differing parameters with the intention of making the function of these systems more easily perceived.

1. BACKGROUND

1.1. Sonification

The choice to attempt to sonify these equations is rooted in sonification theory, primarily the idea that auditory perception in humans is exceptional at recognizing patterns, especially in temporal changes [2]. As the reaction diffusion systems are entirely based upon patterns, it is then logical that an auditory representation may represent them well. The next question then becomes what approach to take in sonifying these equations. Early on in discussion, the similarity between reaction diffusion systems and mechanical string equations was noted, to possibly be used as a link for auditory interpretation. Following this link, it was determined that then, a parameter mapping approach might be apt [3]. The parameter mapping approach would be used by taking normalized derived values from the reaction diffusion systems, and using them to drive an additive synthesis-based sonification system, discussed in more depth later.

1.2. Reaction Diffusion

Turing's reaction diffusion mechanisms are a chemical reaction model which suggests that there are certain conditions under which the reaction and diffusion of chemicals will result in a steady state of an identifiable pattern. The general equation for these mechanisms is as follows. [1]

$$\frac{\delta c}{\delta t} = f(c) + D\nabla^2 c. \tag{1.1}$$

This equation can be generically treated as a partial differential equation, the equation type which we also use to solve the vibration of a string. This is then used as the first link

between the two systems. In the previous equation, our main point of interest will be in the function f(c). Here, c is called the vector of morphogen concentrations, and f is representative of the reaction kinetics. In other words, c represents how much of the chemical is present, and f dictates how it moves to the part of the solution it is present in. Solving a number of further equations for the reaction kinetics in particular yields us three parameters which determine whether a given mode can be present in a certain part of the steady state. These three parameters are given by the following. [1]

$$a = \frac{k_1}{k_2}\left(\frac{k_3}{k_2}\right)^{1/2}, \; b = \frac{k_4}{k_2}\left(\frac{k_3}{k_2}\right)^{1/2}, \; d = \frac{D_B}{D_A} \tag{1.2}$$

These equations are given without the context for how they are solved, but the main thing to understand is that a and b regulate the regions of a pattern, while d is the diffusivity constant, how the reaction moves in the region. When these three values are solved for, it can be determined whether or not a mode is allowed in a region, and thus, whether or not it will be expressed in the final pattern. These values are generically dimensionless, applied then to chosen dimensions when running a simulation. When running the MATLAB simulations, these parameters are chosen at the beginning, and used to solve for the presence of each mode. Each mode, if present, then has its own associated eigenvalue, and wavenumber, k, upon solving the original equation for the space given by a, b, and d.

1.3. Connecting to String Equations

The link between string equations and the previously discussed reaction diffusion system lies in the present modes, their eigenvalues, and the k values from the linear stability analysis. It may be noted that the two-dimensional domain systems as discussed more closely resemble a vibrating plate or membrane. However, if the domain is changed to be one dimension, it much more closely resembles a string. It can be additionally noted that the modes of the reaction diffusion system in two dimensions in square and rectangular domains are symmetric, and this is reflected in their derived values. This is useful when we transition to additive synthesis for these simple shape domains, because it means that in a case of present modes (0, 1) and (1, 0), these modes will have the same derived values, and as such can be condensed to a single overtone. All of this to say, the representation of a two-dimensional equation in what seems to be a one-dimensional mapping is not arbitrary, but rather, a conceptual simplification without losing any information.

String equations, like the reaction diffusion equations, have a number of present modes, sometimes referred to as harmonics or overtones, which are derived from a number of initial conditions and have individual contributions to the final wave shape [4]. As string equations function on the idea of the complex waveform being decomposed into sinusoids, we will be doing the opposite, using additive synthesis with sinusoids. It then follows that we can apply our mapping somewhat directly, understanding the direct match between modes, keeping in mind their symmetry, eigenvalues being a representation of contribution, and k values representing a wavenumber. The process of mapping will be explained further in the system design.

1.4. Additive Synthesis

Where the string equations decompose complex waveforms into individual sinusoids through Fourier transforms, additive synthesis composes complex waveforms by adding together individual sinusoids. Additive synthesis is particularly useful for our purpose as the parameters that we have describe individual modes, which can be translated to the string equations as sinusoids with some relation to the first mode. We will use this technique to create mappings with particular timbral qualities determined by the derived values from reaction diffusion systems.

2. SYSTEM DESIGN

2.1. Requirements

To fully explore the different stages in complexity of reaction diffusion equations, it is proposed to have multiple mappings for 1) non-growing square domains, 2) growing square to rectangle domains, and 3) a single mapping for a growing limb simulation. Each of these mappings must have both the auditory mapping, as well as a visual representation of the simulation. These mappings will be created using the Max/MSP programming environment [4].

2.2. Obtaining Simulation Data

To obtain data and images for simulations, MATLAB code was provided. This code would allow for any number of chosen parameters to be selected and then a simulation generated with them. Initial parameter values could be chosen for a, b, and d, (as previously discussed in section 1.2) which would then be used to run the set of equations with a given simulation size (also chosen by the user). For non-growing domains, the output would result in a .png file as a visual representation of the reaction's concentrations at each given time step. Playing all of these time steps one after another results in the simulation animation. The non-growing domain code would also provide a single linear stability analysis file which would provide the present modes and their relevant eigenvalues and k values. For growing domain simulations, there exists the same pipeline for .png file generation. Additionally, there is now a linear stability analysis file provided for each time step. This allows for our mapping to adjust for modes present at individual time steps.

2.3. Displaying Simulation Animations

It was decided for the purposes of workflow that it would be best to animate the simulations alongside the audio synthesis. This way, alterations to code handling audio synthesis could be observed in real time alongside the video to ensure their cohesion. Therefore, in order to keep the video and audio portions together, we used Jitter [6] in the Max environment. Two main patches were created to facilitate this approach. The first script would take a folder of numbered .png files and iterate over them, converting each to a numbered jitter binary file. This was done so that the process of loading image files into a Jitter window would be much less CPU intensive. After this patch completed its work, a second patch, loaded as a subpatch in the production patches, would go through the process of animating the images, playing them one after another in time with the audio synthesis. This patch utilizes a duration in seconds for both the animation and audio to take to reach completion, and then divides this time by the number of frames available for the animation in order to drive a metro object which sets the timing for each frame. The frames are then displayed in a typical fashion for the jit.matrix object. An example image of this second patch is provided below.

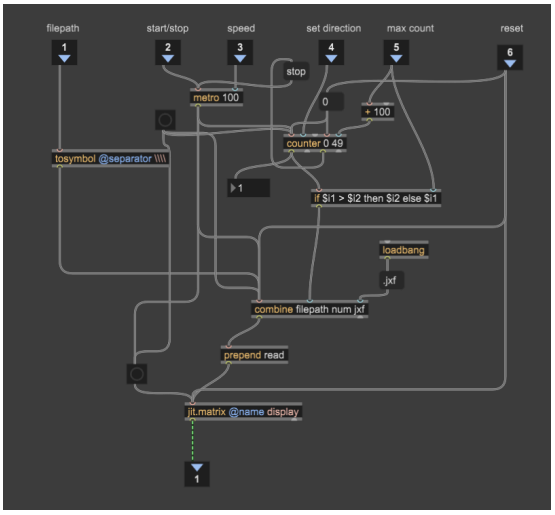

Fig 1. Animation Display Subpatch

2.4. Synthesizing the Data

The main patch driving the additive synthesis of the data is one that sets individual overtone values over time. In a given production patch, there are multiple subpatches which set the overtones individually. These are then all combined together for the final output to complete the additive synthesis process. It is

perhaps best to explain the process of this patch in reverse. What the patch does is ramp a multiplier against a fundamental frequency from 0 to a given wavenumber between 0 and 1. So starting from the end of the patch, we have our wavenumber, the k value from our linear stability analysis, which is simply multiplied against the fundamental frequency to calculate the overtone. However, as mentioned, the k value is between 0 and 1, so we use this normalization to, as a baseline, simply add 1 to it so it will be a value higher than the fundamental. The octave jump of the overtone is determined by the modes present. We simply take the lower number of the mode definition and add it to the normalized and shifted k value. For example, if we have mode (0, 1), then the k value gets 0 added to it, whereas if we have mode (2, 1) it will have 1 added to it as 1 is the smaller mode value. This mapping choice does not necessarily have a strict mathematical justification, and is much more informed by the human perception [2]. This all explains how the overtone for the end of the simulation is calculated, now we can explain how the change over time is calculated. The emergence of the modes over time in the reaction diffusion equations tends to be exponential, with their rate determined by their eigenvalue. As such, we will essentially be modeling our overtone as $x^a * k$ where a is our set eigenvalue, k is the k value set previously, and x is a value between 0 and 1 which should increase linearly. To have x increase in this way, we use a counter object driven by a metronome object (to sync with the video as previously mentioned). The counter counts from 0 to 1000, and is then shifted such that the values that go into our equation increase by 0.001 for 1000 steps, ending at a value of 1. This allows for the overtone's frequency value to ramp exponentially over time, matching with the simulations. Our final use of the eigenvalue as a representation of contribution is for setting the amplitude of the overtone. We use a ratio of the mode's eigenvalue against the maximum eigenvalue of the system to be the ratio of the mode's amplitude compared to the maximum amplitude of the fundamental mode. It was also decided to multiply this value by 0.75 of the maximum amplitude, again considering the perception of the human, where most instruments that we are familiar with do not have their next most contributing mode to be the same as the fundamental. An example image of the synthesis patch is provided below.

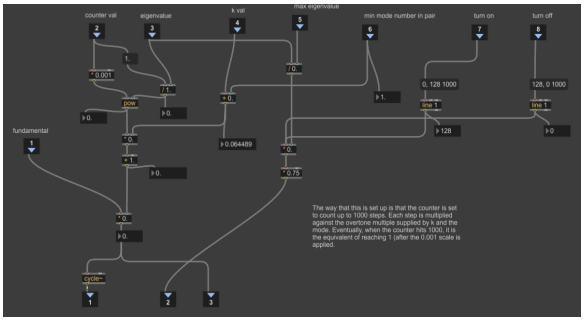

Fig 2. Overtone Calculation Subpatch

Finally, we how the non-growing domain production patches change to become the growing domain patches. They are entirely the same up until this point of discussion, both make use of the same overtone calculation subpatches. The main difference then, is that the growing domains do not always have the same modes present at all points of the simulation. A rather manual process was applied to complete this mapping. Linear stability analysis files for predetermined time intervals were observed, and it was notes which modes were present and not present at these times. Then, a simple if/then structure was applied to increase or decrease the amplitude of the individual modes as they became present or not present at the given time. The overtones ramp continuously, and their amplitude is governed by the time steps and their presence in the simulation. An example image of this structure is provided below.

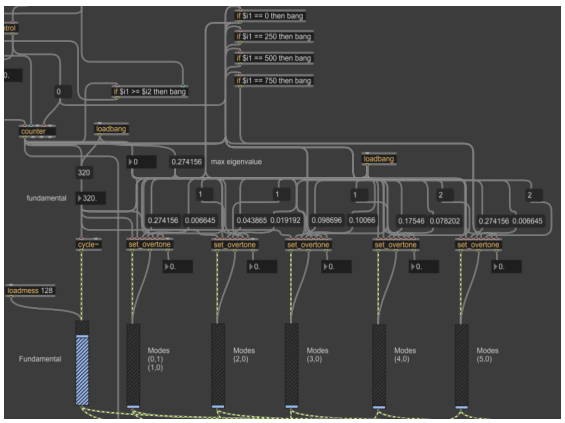

Fig 3. Growing Domain Control Flow

The final mapping decision added to the growing domains is the inclusion of reverb whose delay time grows over time, to simulate the space of the system growing over time. The cverb~ object was used for this purpose, with the delay attribute increasing as the counter increased.

3. RESULTS

This work culminated to create two mappings of non-growing simulations with square domains. One simulation showcased the presence of only three modes, while the other showcased over sixteen. There were also two mappings created for growing domains, these with a similar number of present modes. The first has six present modes over the course of the simulation, while the second has seven. There was one final mapping which was applied to an axolotl limb simulation. However, the intricacies of this equation would have been too difficult to find direct matches for in the time for this project to be completed, so an approximate growing domain with squares and rectangles was created to synchronize with the limb simulation. These videos can all be accessed at the following link: https://drive.google.com/drive/folders/1zKBiFtUpKHWW0Z5B44AlVemgvUe83cwt?usp=sharing.

4. DISCUSSION

This cross-disciplinary work opened a number of different discussions regarding the initial research for which more work would be required to investigate further. One major discussion point was in the definition of homogeneity in the reaction diffusion equation. Before the reaction occurs, the modeled solution is said to be in a state of homogeneity. However, the representation of this state is not necessarily understood. It mostly is not understood whether this state should be modeled as more static or more nonstatic. Following this observation, it was a point of debate whether additive or subtractive synthesis then should have been chosen for the sonification. The subtractive synthesis would have begun from a starting point of white noise, as a representation of homogeneity as more nonstatic. However, the starting point of a single sinusoid was deemed more fitting, because noise is intrinsically dynamic, which the state of homogeneity would not be described as. However, this is still a question that could be open for more interpretations. Another aspect of the original research that this work sought to explore was the open question of biological processes that happen in the transitions of the growing stages. It is largely unknown when exactly certain modes appear or disappear, as well as whether or not this timing is strictly relevant to limb development. Our sonification made an attempt to provide a representation of what might be happening, but there is not currently a way to map this directly, so this will remain a best guess until further research is completed. Finally, there is the general question of whether or not sonification as an approach was useful for the understanding of the underlying research, both mathematically and biologically. The theory would suggest that it is a good match, and previous showcases of this work have yielded positive feedback. However, this is a largely subjective question, especially since the broader intention is simply education. We have generally found that the comparison to aspects of audio generation that are understood in an everyday context, such as the timbre of

instruments has allowed this project to be generally understood by lay audiences, without going into extreme detail for the underlying mathematics. We believe then, that the project has been generally successful in its original intention.

5. REFERENCES

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